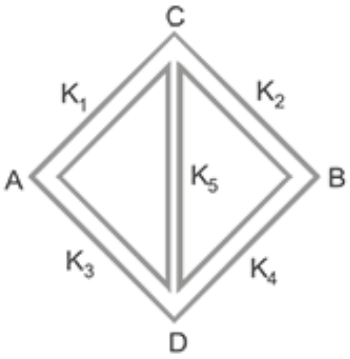
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Five rods of same dimensions are arranged as shown in the figure. They have thermal conductivities $\mathrm{K}_1,\ \mathrm{K}_2,\ \mathrm{K}_3,\ \mathrm{K}_4$ and K_5. When points A and B are maintained at different temperatures, no heat flows through the central rod if



(A) $\mathrm{K}_1 = \mathrm{K}_4$ and $\mathrm{K}_2 = \mathrm{K}_3$

(B) $\mathrm{K}_1\ \mathrm{K}_4 = \mathrm{K}_2\ \mathrm{K}_3$

(C) $\mathrm{K}_1\ \mathrm{K}_2 = \mathrm{K}_3\ \mathrm{K}_4$

(D) $\frac{\mathrm{K}_1}{\mathrm{K}_4} = \frac{\mathrm{K}_2}{\mathrm{K}_3}$

Q.2

Ice starts forming in a lake with water at $0^{\circ}\mathrm{C}$ when the atmospheric temperature is $-10^{\circ}\mathrm{C}$. If the time taken for the first $1\ \mathrm{cm}$ of ice to be formed is 7 hours, then the time taken for the thickness of ice to change from $1\ \mathrm{cm}$ to $2\ \mathrm{cm}$ is

(A) 7 hours

(B) 14 hours

(C) 21 hours

(D) 3.5 hours

Q.3

Two rods of the same length and diameter having thermal conductivities K_1 and K_2 are joined in parallel. The equivalent thermal conductivity of the combination is

(A) $\frac{K_1 K_2}{K_1 + K_2}$

(B) $K_1 + K_2$

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(C) $\frac{K_1+K_2}{2}$

(D) $\sqrt{K_1K_2}$

Q.4

Two rods of same length and area of cross-section $\mathrm{A_1}$ and $\mathrm{A_2}$ have their ends at the same temperature. If $\mathrm{K_1}$ and $\mathrm{K_2}$ are their thermal conductivities, $\mathrm{c_1}$ and $\mathrm{c_2}$ are their specific heats and $\mathrm{d_1}$ and $\mathrm{d_2}$ are their densities, then the rate of flow of heat is the same in both the rods if

(A) $\frac{A_1}{A_2} = \frac{-k_1}{k_2}$

(B) $\frac{A_1}{A_2} = \frac{k_1c_1d_1}{k_2c_2d_2}$

(C) $\frac{A_1}{A_2} = \frac{k_2c_1d_1}{c_2d_2k_1}$

(D) $\frac{A_1}{A_2} = \frac{k_2}{k_1}$

Q.5

Two spheres of different materials one with double the radius and one-fourth wall thickness of the other are filled with ice. If the time taken for complete melting of ice in the larger sphere is 25 minute and for smaller one is 16 minute, the ratio of thermal conductivities of the materials of larger spheres to that of smaller sphere is

(A) $4:5$

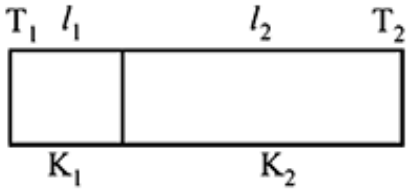
(B) $5:4$

(C) $25:8$

(D) $8:25$

Q.6

One end of a thermally insulated rod is kept at a temperature T_1 and the other at T_2. The rod is composed of two sections of length l_1 and l_2 and thermal conductivities K_1 and K_2 respectively. The temperature at the interface of the two sections is



(A) $\frac{(K_1l_1T_1+K_2l_2T_2)}{(K_1l_1+K_2l_2)}$

(B) $\frac{(K_2l_2T_1+K_1l_1T_2)}{(K_1l_1+K_2l_2)}$

(C) $\frac{(K_2l_1T_1+K_1l_2T_2)}{(K_2l_1+K_1l_2)}$

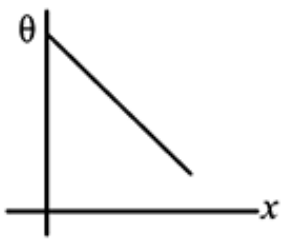
(D) $\frac{(K_1l_2T_1+K_2l_1T_2)}{(K_1l_2+K_2l_1)}$

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Q.7

A long metallic bar is carrying heat from one of its ends to the other end under steady-state. The variation of temperature θ along the length x of the bar from its hot end is best described by which of the following figures?

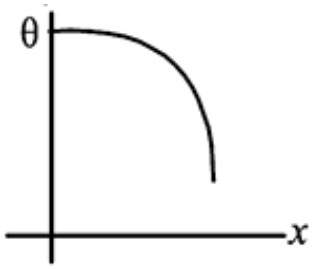
(A)



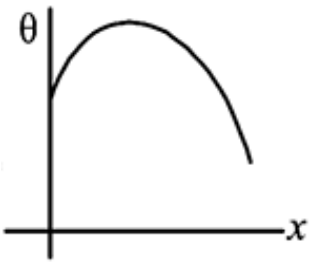
(B)



(C)



(D)



Q.8

Two rods of same length and transfer a given amount of heat in 12 second, when they are joined as shown in figure (i). But when they are joined as shwon in figure (ii), then they will transfer same heat in same conditions in Fig. (ii)

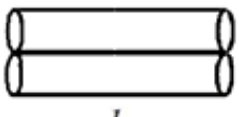


Fig. (i)

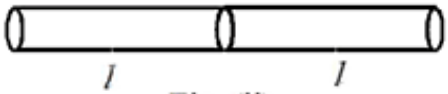


Fig. (ii)

(A) 24 s

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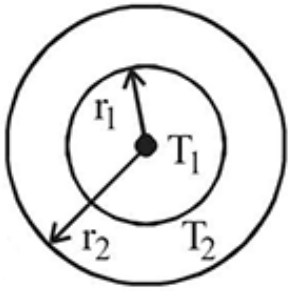
(B) 13 s

(C) 15 s

(D) 48 s

Q.9

The figure shows a system of two concentric spheres of radii r_1 and r_2 are kept at temperatures T_1 and ${}^2T_2$ respectively. The radial rate of flow of heat in a substance between the two concentric spheres is proportional to



(A) $\ell n\left(\frac{r_2}{n}\right)$

(B) $\frac{(r_2-r_1)}{(r_1r_2)}$

(C) (r_2-r_1)

(D) $\frac{r_1r_2}{(r_2-r_1)}$

Q.10

A wall has two layers A and B made of different materials. The thickness of both layers is the same. The thermal conductivity of A and B are K_A and K_B such that $K_A = 3K_B$. The temperature across the wall is $20^\circ\mathrm{C}$. In thermal equilibrium

(A) the temperature difference across A is $15^\circ\mathrm{C}$

(B) the temperature difference across A is $5^\circ\mathrm{C}$

(C) the temperature difference across A is $10^\circ\mathrm{C}$

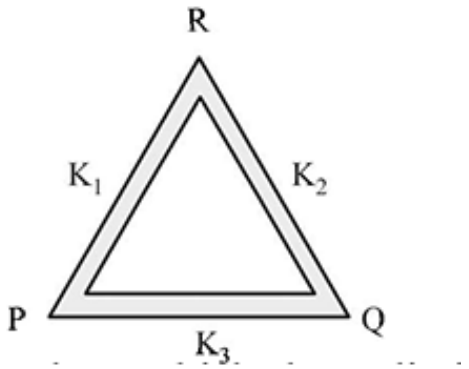
(D) the rate of transfer of heat through A is more than that through B

Q.11

Three rods of same dimensions are arranged as shown in figure, have thermal conductivities K_1, K_2 and K_3. The points P and Q are maintained at different temeratures for the heat to flow at the same rate along PRQ and

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PQ. Then which of the following option is correct?



(A) $K_3 = \frac{1}{2}(K_1 + K_2)$

(B) $K_3 = K_1 + K_2$

(C) $K_3 = \frac{K_1 K_2}{K_1 + K_2}$

(D) $K_3 = -2(K_1 + K_2)$

Q.12

If two rods A and B of equal length L, and different areas of cross-section A_1 and A_2 have one end each at temperature T_1 and T_2, have equal rates of flow of heat, then

(A) $A_1 = A_2$

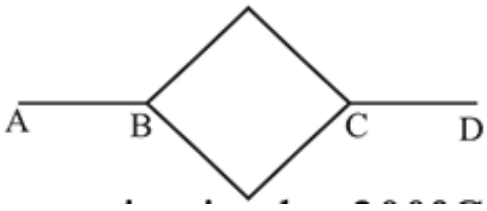
(B) $\frac{A_1}{A_2} = \frac{K_1}{K_2}$

(C) $\frac{A_1}{A_2} = \frac{K_2}{K_1}$

(D) $K_1 = K_2$

Q.13

Six identical conducting rods are joined as shown in figure. Points A and D are maintained at $200^{\circ}\mathrm{C}$ and $20^{\circ}\mathrm{C}$ respectively. The temperature of junction B will be



(A) $120^{\circ}\mathrm{C}$

(B) $100^{\circ}\mathrm{C}$

(C) $140^{\circ}\mathrm{C}$

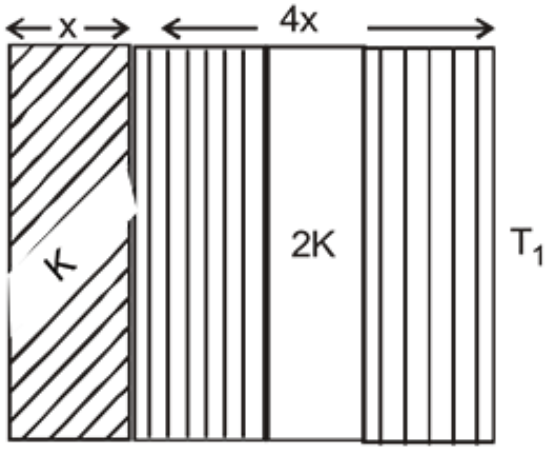
(D) $80^{\circ}\mathrm{C}$

Q.14

The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients

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of thermal conductivity K and $2K$ and thickness x and $4x$, respectively, are T_2 and T_1 $(T_2 > T_1)$. The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2-T_1)K}{x}\right) f$, with f equal to



(A) $\frac{2}{3}$

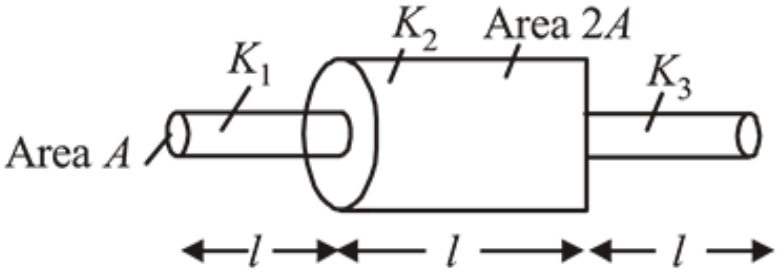
(B) $\frac{1}{2}$

(C) 1

(D) $\frac{1}{3}$

Q.15

What is the equivalent thermal conductivity of the rods in figure given below, if the length of each cylinder be ℓ and area of cylinder having thermal conductivities K_1 and K_3 be A while that of the middle cylinder having thermal conductivity K_2 be 2 A?



(A) $2\left[\frac{1}{K_1}+\frac{1}{2K_2}+\frac{1}{K_3}\right]$

(B) $\frac{1}{\left[\frac{1}{K_1}+\frac{1}{2K_2}+\frac{1}{K_3}\right]}$

(C) $\frac{2}{5\left[\frac{1}{K_1}+\frac{1}{2K_2}+\frac{1}{K_3}\right]}$

(D) None of these

Q.16

One end of a uniform rod of length 1 m is placed in boiling water while its other end is placed in melting ice. A point P on the rod is maintained at a constant temperature of 800°C. The mass of steam produced per second is equal to the mass of ice melted per second. If specific latent heat of steam is 7 times the specific latent heat of ice, the distance of P from the steam chamber must be

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(A) $(1/7)\mathrm{m}$

(B) $(1/8)\mathrm{m}$

(C) $(1/9)\mathrm{m}$

(D) $(1/10)\mathrm{m}$

Q.17

Which of the following circular rods (given radius r and length l), each made of the same material and whose ends are maintained at the same temperature will conduct most heat?

(A) $\mathrm{r} = \mathrm{r}_0; \mathrm{l} = \mathrm{l}_0$

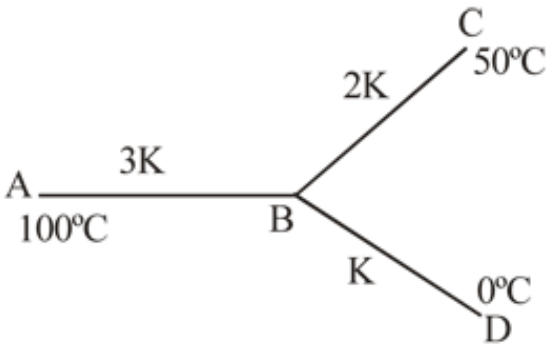
(B) $\mathrm{r} = 2\mathrm{r}_0; \mathrm{l} = \mathrm{l}_0$

(C) $\mathrm{r} = \mathrm{r}_0; \mathrm{l} = 2\mathrm{l}_0$

(D) $\mathrm{r} = 2\mathrm{r}_0; \mathrm{l} = 2\mathrm{l}_0$

Q.18

Three rods of same dimensions have thermal conductivities $3\ \mathrm{K}, 2\ \mathrm{K}$ and K. They are arranged as shown, with their ends at $100^\circ\mathrm{C}, 50^\circ\mathrm{C}$ and $0^\circ\mathrm{C}$. The temperature of their junction is



(A) $75^\circ\mathrm{C}$

(B) $(200/3)^\circ\mathrm{C}$

(C) $40^\circ\mathrm{C}$

(D) $(100/3)^\circ\mathrm{C}$

Q.19

A uniform metal ring with centre C have two points A and B, such that angle ACB is θ. A and B are maintained at two different constant temperatures. If the angle between A and B, i.e., $\theta = 180^\circ$, the rate of heat flow from A to B is $1.2W$, then what will be the rate, when $\theta = 90^\circ$

(A) $0.6\ \mathrm{W}$

(B) $0.9\ \mathrm{W}$

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(C) $1.6\ \mathrm{W}$

(D) $1.8\ \mathrm{W}$

Q.20

The ends of the two rods of different materials with their lengths, diameters of cross-section and thermal conductivities all in the ratio $1:2$ are maintained at the same temperature difference. The rate of flow of heat in the shorter rod is $1\mathrm{cal/s}$. Wat is the rate of flow of heat in the larger rod?

(A) $1\mathrm{cals}^{-1}$

(B) $4\mathrm{cals}^{-1}$

(C) $8\mathrm{cals}^{-1}$

(D) $16\mathrm{cals}^{-1}$

Answers & Solutions

Q.1 Answer:

$\mathrm{K_1\ K_4 = K_2\ K_3}$

Solution:

As we know that thermal resistance is given as: $\mathrm{R} = \frac{\mathrm{L}}{\mathrm{KA}}$

So, simply we apply the concept of balanced wheat-stone bridge for which no current will flow across the central rod,and mathematically it is given as:

$$\mathrm{R_1R_4 = R_2R_3}$$

Here, ratio of $\frac{\mathrm{L}}{\mathrm{A}}$ **is constant.**

Hence, we get the following relation:

$$\mathrm{K_1\ K_4 = K_2\ K_3}$$

Q.2 Answer:

21 hours

Solution:

Let us assume that x **length of ice has become formed, and to form a small strip of ice of length** dx**, dt time is required.**

then

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$\frac{dQ}{dt} = \frac{KA\Delta T}{x}$

$\frac{d(mL)}{dt} = \frac{KAAT}{x}$

$\frac{A\rho_w Ldx}{dt} = \frac{KA\Delta T}{x}$

$dt = \frac{\rho_w Lxdx}{K\Delta T}$

where K is thermal conductivity of ice, A is area of the lake, L is latent heat of ice, m and ρ_w are mass density of ice and ΔT is the temperature difference. by integrating the above equation on both sides

$\int_0^t dt = \frac{\rho_w L}{K\Delta T}\int_0^1 xdx$

$t = \frac{\rho_w L}{K\Delta T}\frac{1^2}{2}$

$t = C\frac{1^2}{2}$

where $C = \frac{\rho_w L}{K\Delta T}$

Given that to form 1 cm thickness of ice, 7hours time has taken then time required to form 2 cm thickness of ice is given by

$\frac{t_1}{1_1^2} = \frac{t_2}{1_2^2}$

$t_2 = \frac{t_1}{1_1^2}1_2^2$

$= \frac{7}{1} \times 4 \text{ hours}$

$= 28 \text{ hours}$

time taken to form ice from 1 cm to 2 cm is

$t_2 - t_1 = 28 - 7 = 21 \text{ hours}$

option (C) is the correct answer

Q.3 Answer:

$\frac{K_1+K_2}{2}$

Solution:

In parallel combination, the equivalent thermal conductivity is given by

$K = \frac{K_1A_1+K_2A_2+K_3A_3+\ldots\ldots+K_nA_n}{A_1+A_2+A_3+\ldots\ldots+A_n}$

For two rods of equal area, $K = \frac{(K_1+K_2)A}{2A}$ **(if** $A_1 = A_2 = A$**)**

$\Rightarrow K = \frac{K_1+K_2}{2}$

Q.4 Answer:

$\frac{A_1}{A_2} = \frac{k_2}{k_1}$

Solution:

$\frac{Q}{t} = K_1\ A_1\frac{d\theta}{dx} = K_2\ A_2\frac{d\theta}{dx}$

Q.5 Answer:

$8 : 25$

Solution:

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Radius of small sphere $=\mathrm{r}$ Thickness of small sphere $=\mathrm{t}$ Radius of bigger sphere $=2\mathrm{r}$ Thickness of bigger sphere $=\mathrm{t}/4$ Mass of ice melted $=$ (volume of sphere) $\times$ (density of ice)

Let K_1 and K_2 be the thermal conductivities of larger and smaller sphere. For bigger sphere,

$\frac{\mathrm{K}_1 4\pi(2\mathrm{r})^2\times 100}{\mathrm{t}/4}=\frac{\frac{4}{3}\pi(2\mathrm{r})^3\rho\mathrm{L}}{25\times 60}$

For smaller sphere,

\frac{\mathrm{K}_{2} \times 4 \pi \mathrm{r}^{2} \times 100}{\mathrm{t}}=\frac{\frac{4}{3} \pi \mathrm{r}^{3} \mathrm{\rhoL}}{16 \times 60}

$\therefore \frac{\mathrm{K}_1}{\mathrm{K}_2}=\frac{8}{25}$

$$\therefore \frac{\mathrm{K}_1}{\mathrm{K}_2}=\frac{8}{25}$$

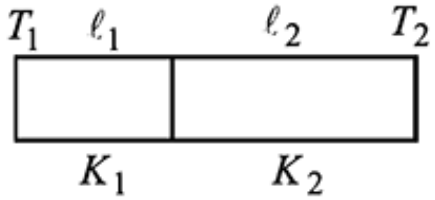
Q.6 Answer:

$\frac{(K_1 l_2 T_1+K_2 l_1 T_2)}{(K_1 l_2+K_2 l_1)}$

Solution:

Let T be the temperature of the interface. As the two sections are in series, the rate of flow of heat in them

will be equal.



$\therefore \frac{K_1 A(T_1-T)}{\ell_1}=\frac{K_2 A(T-T_2)}{\ell_2}$

where A is the area of cross-section.

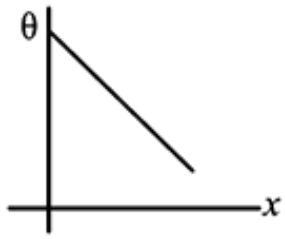
or, $\quad K_1 A\left(T_1-T\right) \ell_2=K_2 A\left(T-T_2\right) \ell_1$

or, $\quad K_1 T_1 \ell_2-K_1 T \ell_2=K_2 T \ell_1-K_2 T_2 \ell_1$

or, $\left(K_2 \ell_1+K_1 \ell_2\right) T=K_1 T_1 \ell_2+K_2 T_2 \ell_1$

$\therefore T=\frac{K_1 T_1 \ell_2+K_2 T_2 \ell_1}{K_2 \ell_1+K_1 \ell_2}=\frac{K_1 \ell_2 T_1+K_2 \ell_1 T_2}{K_1 \ell_2+K_2 \ell_1}$

Q.7 Answer:



Solution:

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The heat flow rate is given by $\frac{dQ}{dt} = \frac{kA(\theta_1 - \theta)}{x}$
$\Rightarrow \theta_1 - \theta = \frac{x}{kA}\frac{dQ}{dt} \Rightarrow \theta = \theta_1 - \frac{x}{kA}\frac{dQ}{dt}$
where θ_1 **is the temperature of hot end and** θ **is temperature at a distance** x **from hot end. The above equation can be graphically represented by option (a).**

Q.8 Answer:

48 s

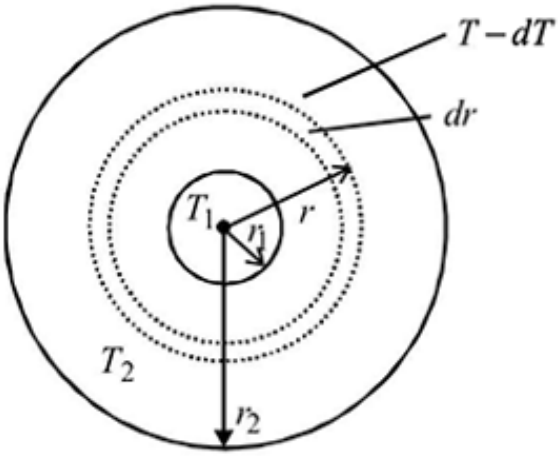
Solution:

$t \propto \frac{\ell}{\mathrm{A}} t' \propto \frac{2\ell}{\mathrm{A}/2}$
$\frac{\mathrm{t}'}{\mathrm{t}} = 4\frac{\ell/\mathrm{A}}{\ell/\mathrm{A}}$
$\mathrm{t}' = 4 \times \mathrm{t}$
$3/\mathrm{t}' = 48$ s

Q.9 Answer:

$\frac{r_1 r_2}{(r_2 - r_1)}$

Solution:



Consider a shell of thickness (dr) **and of radiius** (r) **and let the temperature of inner and outer surfaces of this shell be** T **and** $(T - dT)$ **respectively.** $\frac{dQ}{dt}$ **= rate of flow of heat through it** $= \frac{KA[(T-dT)-T]}{dr} = \frac{-KAdT}{dr}$
$= -4\pi K r^2 \frac{dT}{dr} \quad (\because A = 4\pi r^2)$
To measure the radial rate of heat flow, integration technique is used, since the area of the surface through which heat will flow is not constant. Then, $\left(\frac{dQ}{dt}\right)\int_n^{r_2} \frac{1}{r^2} dr = -4\pi K \int_{T_1}^{T_2} dT$
$\frac{dQ}{dt}\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = -4\pi K\left[T_2 - T_1\right]$
or $\frac{dQ}{dt} = \frac{-4\pi K r_1 r_2 (T_2 - T_1)}{(r_2 - r_1)}$
$\therefore \quad \frac{dQ}{dt} \propto \frac{r_1 r_2}{(r_2 - r_1)}$

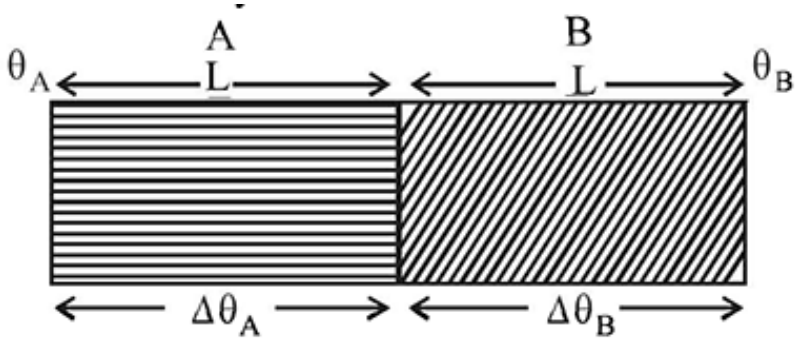
Q.10 Answer:

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the temperature difference across A is $5^{\circ}\mathrm{C}$

Solution:

The wall of two layers A and B are connected in series. Then, heat flowing per second across both wall layers are same i.e



$\mathrm{H}_1 = \mathrm{H}_2$

$\frac{\mathrm{K_A A}\Delta\theta_\mathrm{A}}{\mathrm{L}} = \frac{\mathrm{K_B A}\Delta\theta_\mathrm{B}}{\mathrm{L}} \Rightarrow \mathrm{K_A}\Delta\theta_\mathrm{A} = \mathrm{K_B}\Delta\theta_\mathrm{B}$

$\Rightarrow 3\,\mathrm{K_B}\Delta\theta_\mathrm{A} = \mathrm{K_B}\Delta\theta_\mathrm{B} \quad (\text{ as } \mathrm{K_A} = 3\,\mathrm{K_B})$

$\Rightarrow 3\Delta\theta_\mathrm{A} = \Delta\theta_\mathrm{B}$

Also $\Delta\theta_\mathrm{A} + \Delta\theta_\mathrm{B} = 20^{\circ}\mathrm{C}$

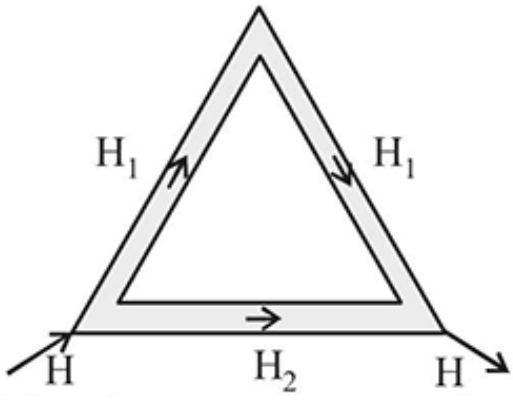
From (i) \& (ii) $\therefore \Delta\theta_\mathrm{A} + 3\Delta\theta_\mathrm{A} = 20^{\circ}\mathrm{C} \Rightarrow 4\Delta\theta_\mathrm{A} = 20^{\circ}\mathrm{C}$

$\Rightarrow \Delta\theta_\mathrm{A} = 5^{\circ}\mathrm{C}$

Q.11 Answer:

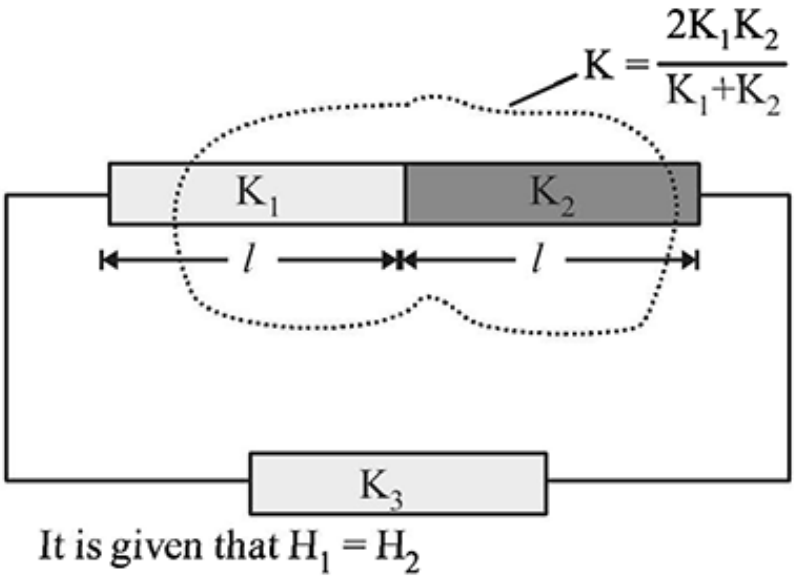
$K_3 = \frac{K_1 K_2}{K_1 + K_2}$

Solution:



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The given arrangement of rods can be redrawn as follows



It is given that $H_1 = H_2$

It is given that $\mathrm{H}_1 = \mathrm{H}_2 \Rightarrow \frac{KA(\theta_1-\theta_2)}{2l} = \frac{K_3A(\theta_1-\theta_2)}{l} \Rightarrow K_3 = \frac{K}{2} = \frac{K_1K_2}{K_1+K_2}$

Q.12 Answer:

$\frac{A_1}{A_2} = \frac{K_2}{K_1}$

Solution:

$$\frac{\Delta Q_1}{\Delta t} = \frac{\Delta Q_2}{\Delta t}$$

$$\Rightarrow K_1A_1\frac{T_1-T_2}{L_1} = K_2A_2\frac{T_1-T_2}{L_2}$$

$$(L_1 = L_2) \quad \therefore \frac{A_1}{A_2} = \frac{K_2}{K_1}$$

Q.13 Answer:

$140^\circ\mathrm{C}$

Solution:

Temperature difference between the end points A **and** $\mathrm{D} = 200 - 20 = 180^\circ\mathrm{C}$

As the resistances for the three parts are equal, the temperature difference must be distribuited equally in the three parts $(= 180/\ 3 = 60^\circ\mathrm{C})$

$\therefore$ **Temperature of** $\mathrm{B} = 200^\circ\mathrm{C} - 60^\circ = 140^\circ\mathrm{C}.$

Q.14 Answer:

$\frac{1}{3}$

Solution:

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The thermal resistance is given by

$\frac{x}{KA}+\frac{4x}{2KA}=\frac{x}{KA}+\frac{2x}{KA}=\frac{3x}{KA}$

$\therefore \frac{dQ}{dt}=\frac{\Delta T}{\frac{3x}{KA}}=\frac{(T_2-T_1)KA}{3x}$

$=\frac{1}{3}\left\{\frac{A(T_2-T_1)K}{x}\right\} \quad \therefore f=\frac{1}{3}$

Q.15 Answer:

None of these

Solution:

$\frac{\ell}{K_1A}+\frac{\ell}{2K_2A}+\frac{\ell}{K_3A}=\frac{\ell}{KA}+\frac{\ell}{2KA}+\frac{\ell}{KA}$

$\frac{5}{2K}=\left(\frac{1}{K_1}+\frac{1}{2K_2}+\frac{1}{K_3}\right)$

$K_{\text{eq}}=\frac{5}{2\left(\frac{1}{K_1}+\frac{1}{2K_2}+\frac{1}{K_3}\right)}$

Q.16 Answer:

$(1/9)\text{m}$

Solution:

$Q_1=\text{k}A\left(\frac{800-100}{x}\right)t=mL_{\text{steam}}$

$Q_2=\text{k}A\left(\frac{800-0}{1-x}\right)t=mL_{ice}$

Dividing, $\left(\frac{700}{x}\right)\left(\frac{1-x}{800}\right)=7$

or $1-x=8x$ **or** $x=\frac{1}{9}\ \text{m}$

Q.17 Answer:

$\text{r}=2\text{r}_0;\text{l}=\text{l}_0$

Solution:

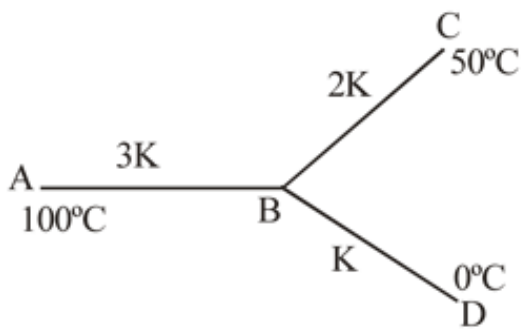
Heat is given by $\text{H}=\text{K}\text{A}\Delta\theta_{\text{x}}=\text{K}\pi\text{r}^2\frac{\Delta\theta}{\text{l}}$

For option B **the value of** H **will be more.**

Q.18 Answer:

$(200/3)^\circ\text{C}$

Solution:



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Let θ be the temperature of the junction (B)

rate of flow of heat through $\mathrm{CB}+$ rate of heat through $\mathrm{DB}=$ rate of heat flow though AB

thermal resistance of the rod $\mathbf{R}=\frac{1}{\mathbf{KA}}$

length and area of the rod is same

solving it $\theta=(200/3)^{\circ}\mathbf{C}$

Q.19 Answer:

$1.6\ \mathrm{W}$

Solution:

Let $\Delta \mathrm{T}=$ temperature difference between the rings.

$\mathrm{R}=$ total thermal resistance of the ring.

When $\theta=180^{\circ}$ the resistances will be $\mathrm{R}/2, \mathrm{R}/2$ and both are in parallel, so equivalent resistance $=\mathrm{R}/4$

Rate of total heat flow $=\frac{\mathrm{Q}_1}{t}=1.2=\frac{\Delta \mathrm{T}}{(\mathrm{R}/4)} \quad 0.3=\frac{\Delta \mathrm{T}}{\mathrm{R}}$

When $\theta=90^{\circ}$ there are two sections, with resistances, $\mathrm{R}/4$ and $3\mathrm{R}/4$ in parallel.

So, equivalent resistance $=\frac{3R}{16}$

Rate of heat flow $=\frac{Q_2}{t}=\frac{\Delta T}{\left(\frac{3R}{16}\right)}$

$=\frac{16}{3}=\left(\frac{\Delta T}{R}\right)$

$=\frac{16}{3} \times 0.3\ \mathrm{W} \quad =1.6\ \mathrm{W}$

Q.20 Answer:

$4 \mathrm{cals}^{-1}$

Solution:

$\left(\frac{Q}{t}\right)_1=\frac{K A\left(\theta_1-\theta_2\right)}{d}=1 \mathrm{cals}^{-1}$

$\left(\frac{\mathrm{Q}}{\mathrm{t}}\right)_2=\frac{2\ \mathrm{K}(4\ \mathrm{A})\left(\theta_1-\theta_2\right)}{2\ \mathrm{d}}$

$=\frac{4 \mathrm{KA}\left(\theta_1-\theta_2\right)}{\mathrm{d}}=4 \mathrm{cals}^{-1}$

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